

THE ARCHITECTURE OF
EMERGENCE

THE EVOLUTION OF FORM IN NATURE AND CIVILISATION

Michael Weinstock

Credits for all black and white drawings

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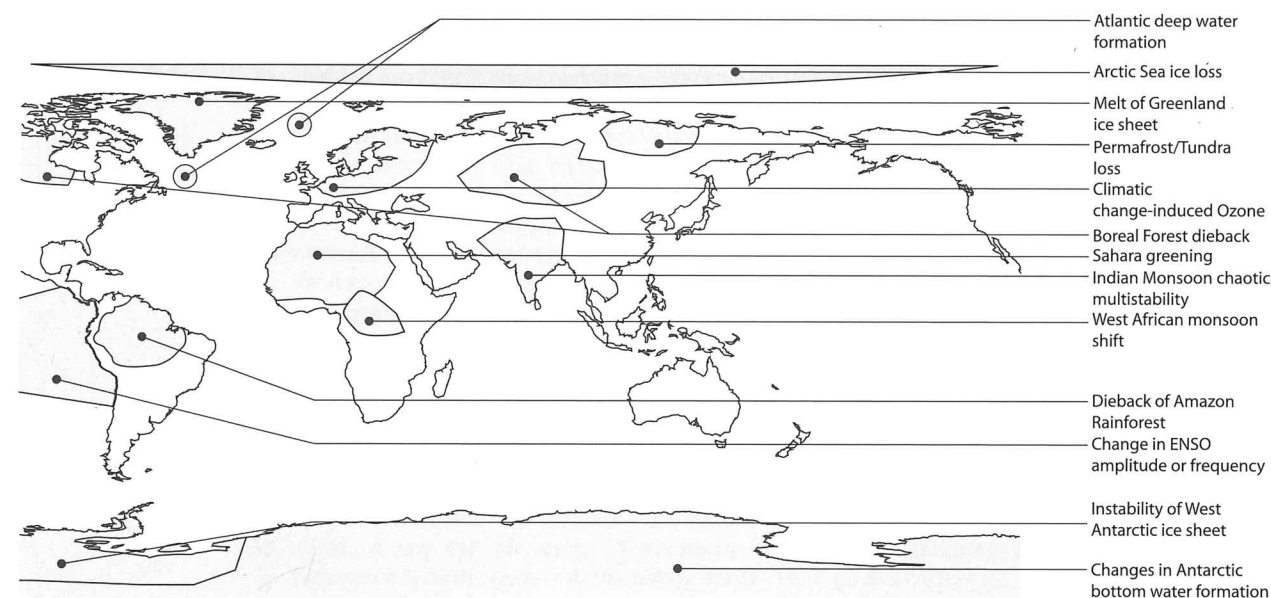
To Diane

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9.1 Critical Thresholds

There are many continental scale ecological and climatic systems that are known to be sensitive to temperature changes, and that are close to their critical threshold and have begun to change now. As the world continues to warm, the transition of one or more system through its critical threshold will provide positive feedback that has the potential to initiate a cascade of changes in others. The large and multiple changes produced by transition through critical threshold are irreversible on millennia timescales.

Emergence

All forms and all systems change over time, and the forms of nature and the cultural forms of civilisation are inextricably entwined. Humans emerged and evolved within nature, and all the forms of human culture have evolved and developed as part of nature and in turn have extensively modified the natural forms of the surface of the earth, the ecological systems of living forms that exist upon it, and the climate 'metasystem'. All the systems of the world, in nature and in civilisation, have tended to evolve until they are poised close to the critical threshold of stability. The outcomes of the collapse and reorganisation of the systems of civilisation have been the complete loss of social order and dispersal of the people by famine, disease, abandonment of cities and mass migration; a simplification into smaller and simpler assemblies or a reorganisation to a lower level of complexity; and the reordering of the cities, people and subsidiary systems into a more complex system with a higher flow of energy and information. The world is within the horizon of systemic change that will cascade through all the systems of nature and civilisation, and new forms will emerge.

The architecture of all the forms of nature, their arrangement of material in space and over time, emerges from the dynamic interaction of energy and material within complex systems. The systems of the climate, the oceans and atmosphere, and the surface of the land also interact with each other, with many exchanges of energy and materials at varying scales of time and space. All the forms of life on the surface of the earth, including humans, have also emerged from the process of complex systems that are coupled to the transmission of biologically encoded information over time. Living forms exist in varied populations, and where they organise themselves into social collectives, culture emerges. Culture provides another system for the transmission of information down through time, linked to, but distinct from, the biological system. Culturally produced forms exist within, and are modified by, all the other natural systems, and as their numbers have increased they in turn have accelerated or inhibited changes to those systems.

All forms change over time. The forms of clouds and storms, of mountains, rivers and deserts are not static things but are constantly being broken down and renewed. The forms of life come into being, flourish and die in each generation,

and although these forms may persist through millennia they are also subject to change. Over time living forms change in response to changes in the climate, in the topography of the land and to changes in the ecological systems within which they and other living forms exist. The culturally produced forms of settlements and cities are also constantly being broken down and rebuilt, and in doing so they are changed. The patterns of energy, information and material that flow through all the forms of the world are subject to many oscillations and perturbations. The flow is modified by 'feedbacks', but occasionally there is such an amplification that the system collapses and is reorganised, and a new form emerges. The new form may be either simpler or more complex. An increase in complexity is usually coupled to an increase in the flow of energy and information through the system, and systems that collapse and revert to a simpler organisation are coupled to a reduced flow of energy and information. Over time, there has been a marked tendency of living systems and of cultural systems towards increasingly higher complexity.

CLIMATE, SURFACE AND LIFE

Climate distributes heat around the world by an intricate choreography of forms and behaviour and in doing so it modulates the exchange of energy and material between all other systems. In turn, climate is affected by the exchanges of energy and material between the ocean, the atmosphere, the land, by all living beings and by the extended metabolic systems of humans. The exchanges occur across multiple scales of dimension and time, so that interactions between the systems are complex and non-linear, subject to fluctuations and variations. These in turn affect the intensity of local weather, and modify the dynamics of the climate system through positive feedback that amplifies change, or through negative feedback that inhibits change. Some changes occur within timescales of hours and days, while other changes take place over decades, hundreds or thousands of years. The metasystem is delicately poised, with many different critical thresholds across multiple scales of space and time, so that a small stimulus at one scale may drive a great and rapid change at another.

The forms of the land emerge from the interaction of tectonic forces uplifting the land from below, and the climatic forces of weathering and erosion breaking down the tectonic forms into fragments and sediments that are carried away and deposited elsewhere. The regional and latitudinal patterns of the climate metasystem generate differentiated regimes of temperature, precipitation and wind, with a range of variation and perturbations in each

regime. Weathering, erosion and deposition are produced by the climate; variations and changes to the climate change the behaviour, duration and force of each process. In turn, geomorphic processes also contribute to the generation of climate feedbacks: changes in the 'feedbacks' of surface albedo, increases in greenhouse gases, and aerosols from dessicated soils and deserts. Many landforms are poised at the critical threshold of change, so that small differences to the ratio or balance between force and resistance, energy and material, will precipitate a rapid change of form.

The emergence of new living forms and diverse species is also strongly coupled to the dynamics of climate and its effects on the surface of the earth. Before life emerged, the sun was weaker than it is today, and the chemistry of the atmosphere and the climate were very different. Cyanobacteria were the first photosynthetic forms of life to emerge; photosynthesis absorbs carbon dioxide and excretes oxygen. Over a billion years oxygen levels rose, the levels of heat-retaining greenhouse gases fell, and temperatures began to fall, precipitating the first ice age.

The smallest and simplest of living forms initiated significant changes to the climate, and in consequence to the forms of the surface of the earth. New regional climatic zones and new physical barriers arise from these dynamics, and these induce migrations, shifts in the range of the species, and isolation from the parent species. Exposure to new and different climatic regimes, topography and ecologies modified the regime of natural selection and new species emerged. All the forms of life exist within ecological and climatic regimes that are poised close to the critical threshold of change, and rapid changes have been a significant factor in all extinction and speciation events. Living forms affect the processes of the atmosphere and oceans, and the geomorphic systems of the surface of the earth.

INFORMATION AND METABOLISM IN LIVING FORMS

All living forms are composed of cells, and each cell carries within it the information for the development of the whole form. The genome encodes the programme for the self-assembly of descendants identical to itself, and so it is passed down through time from one generation to the next. The genome consists of only four different molecules, arranged into many long and complicated sequences, in turn arrayed in distinct modules and groups. One small subset of the genome acts as a feedback on the activation of all the other

genes, accelerating or inhibiting growth in varying sequences and patterns. Many of the regulatory genes are similar across multiple species, incorporated into the system from their earliest ancestors. Changes arise in the genome by 'copy errors' and mutations that shuffle the sequences of genes or repeat some segments, which in turn produce changes to the physical form. No new molecules are needed, and the reconfiguration of existing sequences and repetition produces a higher complexity within the genome. As the regulatory genes control the sequence of growth, small changes to this set can produce variations in the size, shape and number of the repeating modules of the 'body plan' in the living form. Most changes do not offer any significant benefit or are harmful to the living form. Changes that are advantageous are preserved by natural selection, and over time produce better adaptations to changed ecological or climatic circumstances. Every living form emerges from two strongly coupled processes, operating over maximally differentiated time spans: the differential development of cells in the growth of an embryo to an adult form, and the evolution diversification of forms over time. Information is the critical vector between them. Complexity builds over time by a sequence of modifications to existing forms, and from small and simple forms to ever larger and more complex forms. The emergence of the general organisation or 'body plan' of vertebrates, and its subsequent modification into amphibians, reptiles, birds and mammals and eventually humans, occurred in this sequence.

Metabolism is the 'fire of life', the system of all living forms that captures energy and materials from their environment, transforms them and transports them in fluids to every cell, and excretes changed materials as waste back into the environment. It functions through surfaces and branching networks that exhibit identical mathematical parameters in all living forms, from the smallest microbes to giant sequoias, from mice to mammoths.

The metabolic processes of living forms produce changes in their environment, and these changes modify the local regime of natural selection. The organisation of whole ecological systems emerges from the multiple interactions between the metabolic processes of all the forms of life that coexist within it. Many living forms externalise some aspects of their metabolism by material constructions, often collectively arrayed and organised. Constructions such as burrows and nests lessen the load on individual metabolisms within the collective by smoothing out fluctuations in the external temperatures and humidity, storing food and providing strategic access to the home range.

The behavioural complexity of the social insects arises from the collective actions of multiple individuals, each of which has only a very small set of simple programmed behaviours. There is no 'central control, no decision making' individual or group. Each individual insect will have a slightly different threshold of sensitivity to temperature, humidity and chemical stimuli that triggers their actions, and a slightly differing degree of response to them. The collective constructions are permanent, usually centrally located, and their mobility patterns are determined by the spatial pattern of food sources within their range. The collective construction is constantly adjusted by adding or removing material, one grain at a time, to regulate constant temperature, humidity and the ratio of oxygen to carbon dioxide. The intelligent behaviour of the collective as a whole emerges from the interaction of millions of very simple individual actions.

The mobility pattern of the great apes also arises from the spatial pattern of distribution and the seasonality of food plants across the large territory of the group. Their material constructions are temporary, abandoned after a day or two. They are individual constructions, but closely grouped at night for defence. Information in the form of ecological knowledge of their home range, and the expertise needed for nest construction, tool making and use is socially acquired, passed down through the generations from adult to child. It is their cultural system for the transmission of information that modifies and extends the inherited and genetically conserved innate responses to environmental stimuli.

Social order, distributed intelligence and patterns of mobility emerge from the processes of externalised metabolisms. Children are born into an externalised metabolism modified by the ancestral population, and the genes that are linked to these traits are inherited and passed on to the next generation. Genes are transmitted between individuals, but the collective as a whole maintains the ecological inheritance of an extended metabolic system from one generation to the next. This changes the regime of natural selection in the favour of the most successful collectives, and it is their genes that will be conserved.

INFORMATION AND METABOLISM IN HUMAN FORMS

The evolution of human forms and their externalised and extended metabolic system is inseparable from the evolutionary development of their culture system for the transmission of information. Culture acts to transmit complex social and ecologically contextualised information down through the generations, and it

has also tended to increase in complexity over time. In humans, like all other living forms, information and the ability to extract and process energy and materials from their environment are strongly coupled. The human form and culture coevolved during an extended period of extremely rapid variations in the climate and ecology, and the ability to adapt to a variety of climates and ecological conditions was genetically conserved. All people alive today are descendants of a small founding population of anatomically modern humans that emerged in east Africa about 135,000 years ago, and possibly from one woman within that population, the 'Mitochondrial Eve'. Ecological and social information, transmitted down through the generations by the cultural systems of humans, provided the 'programme' or set of instructions for the metabolic, spatial and social organisation of the founding system of civilisation. It enabled humans to regulate their collective metabolisms and thrive in a variety of climatic and ecological territories, to expand their numbers and flow across the surface of the earth.

The founding system of civilisation developed for tens of thousand of years, with gradually accelerating flow of information and energy that enabled expansion of their populations until the system reached its maximum capacity in many locations. Abrupt climate change induced ecological changes and famine, and the collapse of the founding system was precipitated. The flow of surviving migrants to the more ecologically favourable river valleys increased their population density, and cities emerged through a process of nucleation. In the more marginal ecological territories, nomadic systems emerged, and their interaction with cities provided another vector for the linking of cities and the flow of information over greater distances. Cities and their networks of settlements expanded and developed, strongly coupled to the dynamic changes of climate and ecology within which they were situated.

The integration of settlements, cities and colonies into larger and more complex hierarchical systems, linked by maritime and land-based nomadic systems, extended their collective metabolic systems over increasingly larger geographical areas. The flow of energy and materials intensified at a regional scale, and the quantity and complexity of information to regulate the flows acted as a positive feedback to the evolutionary development of more complex information systems. Information grew exponentially in the cities, and flowed out from the cities through the networks of settlements, to the colonies and their networks, and between systems of cities. Complex

information systems arose to account for and to regulate the collection, storage and redistribution of food and fuel, the transformation of raw materials and the manufacture of artefacts and construction of ships and buildings. The acceleration in complexity enabled systems of cities to expand their populations beyond the metabolic constraints of their local and regional territories. Complexity consumes energy and generates information, and each increase in complexity required a further acceleration in the flow of energy and information. The increased demands on the system were met by intensification of production and by expansion over larger territories. Information and metabolism each provide a positive feedback on the development of the other, and on the growth of populations that in turn drove the system to expand yet further. The emergence of the imperial systems that extended their metabolic system over continental scale territories was driven by growth in the kinds of information and its quantity, and by the continual growth of population.

The massive expansion of maritime systems linked the networks of systems of cities in Eurasia and North Africa to the Americas, and moved increasingly large quantities of people, materials and information around the world. The interconnected global system slowly increased the velocity and quantity of flows in most parts of the world, and populations continued to grow. The global system increased in complexity, with rapid increases in populations driving metabolic demands that had begun to surpass the ecological resources of North America, Europe and Asia. The lands available for further territorial expansion were both limited and ecologically more marginal, and began to show rising costs and diminishing returns. Intensification of production to meet the accelerating demands was enabled by the emergence of new forms of information, and the consequent coevolution of new 'high density' energy and information systems. New domains of knowledge, new sciences, and information of all kinds accumulated and grew exponentially. The full transition of energy and information systems, from local to national and then continental scales, took four generations¹ in Europe and North America, and longer elsewhere. A second phase of information and energy development followed, with a somewhat shorter two to three generation deployment. Oil and gas, digital computation, analysis and imaging, each with increasingly interconnected manufacturing and distribution infrastructures drove yet further intensification of production, with greater and more rapid flows of energy and information across continents and oceans.

There are remarkable mathematical similarities in the metabolic distribution networks of living forms and the culturally produced networks for the distribution of information and energy of the global system. Although there is wide variation in their individual components and chemical processes, the metabolic branching networks of all living forms, from the smallest of microbes to the very largest blue whale, exhibit the same topological scaling properties. Networks for information distribution exhibit many similar parameters to the hierarchical branching metabolic networks of living forms,² and a great variety of other culturally produced networks also exhibit comparable 'scale free' power law characteristics. This is most likely to be produced by the way in which the systems grow. Both biological and cultural networks grow continuously by the addition of new nodes or hubs, but these new nodes preferentially attach to nodes that are already well connected. In consequence, the topology of the whole network has only few nodes with a high number of connections that link to all of the other nodes that have progressively smaller numbers of connections. Flow patterns are dominated by the highly connected nodes, through which the maximum volume and velocity of energy, information or material flow. These properties characterise the evolution of biological systems and culturally produced metabolic systems.³ It is important, however, not to confuse the topology diagram with the physical reality. Physical infrastructures are generally massive, constructed from dense materials, with geometry and scale that do not resemble biological networks.

CIVILISATION, ECOLOGY AND CLIMATE

The emergence of cities from the founding system and their subsequent growth, and the subsequent development of the descendant systems of cities, land and maritime nomadic systems, imperial systems and the various phases of the global system were characterised by episodic and irregular expansions and contractions. All developed until they were operating close to the limit of their capacity to support their expanding populations, and to manage the complexity of flows of information, energy and materials through their increasingly extended metabolic systems. Over the last 10,000 years humans have increased their numbers from an estimated 5 million⁴ people spread across the world to significantly more than 6 billion today. All human systems have modified ecological systems and the topography of the earth at a variety of spatial and temporal scales, and in doing so they have altered the exchanges between the surface of the earth and the atmosphere. The dynamic thermal processes of the atmosphere are changed, and in consequence the climatic

regimes, ecological system and topography of the surface of earth are altered in turn. In the last 100 years the growth of the population has been exponential,⁵ and it is widely thought that the total will reach 9 billion within two generations.⁶ It is uncertain that the extended metabolism of the world system can be further accelerated to extract sufficient fuel and food energy, materials and water to support 9 billion people.

The megafauna⁷ were driven to extinction on all continents by human predation and by the additional indirect stress on their habitats induced by human occupation. Land clearance by fire promoted the spread of grasslands, replacing the mature forest.⁸ The gathering of wild grains, nuts, fruit and tubers over tens of thousands of years modified the regime of natural selection to the benefit of those plants and trees favoured by humans. Over tens of thousands of years preferential selection had an impact on the genetic diversity of the ecology,⁹ and initiated the genetic changes for the subsequent domestication of wheat, barley, rice, millet and maize. Agriculture has rearranged the trophic processes of ecological systems, modifying the flow of food energy exclusively to support human needs. Humans harvest the primary production of the products of photosynthesis, and this changes the flow of the whole trophic web of ecological systems and in consequence the variation, distribution and population density of the species. Water flow and the carbon exchange between plants and the atmosphere have also been modified.¹⁰ The clearance of forests for agriculture, and the cutting down of trees for fuel and construction over many thousands of years has resulted in the progressive deforestation over three continents, reducing the transpiration of water up into the atmosphere and altering the storage of water in the soil. It has accelerated the spread of savannah grasslands, and the elimination of tree roots has exposed the land to drought and erosion. It has also changed the albedo of the surface. Systematic intensive agriculture has further accelerated these modifications,¹¹ and contributed to the alteration of river flows and their courses and the silting of deltas. Fuel for domestic uses and for the transformation of materials in manufacturing has added carbon dioxide and other greenhouse gases to the atmosphere, together with soot or black carbon particulates, enabling it to retain more heat energy.

Anatomically modern humans evolved in an extended period of multiple climatic and ecological changes. There have been several major episodes of abrupt change and multiple smaller oscillations since then, many with a very rapid onset

of a decade or two. Non-linear perturbations and variations in the monsoon belt tend to induce abrupt cooling in the Atlantic Ocean, and changes to the timing and geographical distribution of rainfall. Changes to the vegetation cover of the land surfaces by human clearance of the forests and woodlands have acted as positive feedback to the changes in temperatures and in the global pattern of precipitation that have varied in duration from decades to several centuries. In many episodes of climatic and ecological changes the agricultural capacity of soils already exhausted by intensive cultivation and overgrazing was further reduced. Energetic returns of extended metabolic systems were diminished, populations declined, and prolonged famines and warfare induced waves of migration and abandonment.

Climate change is strongly coupled to the emergence of all of the cultural systems of civilisation, to their subsequent evolutionary diversifications and developments, expansions and contractions, and to their eventual collapse and reorganisation.

CRITICAL THRESHOLDS

All systems that develop over time pass through a number of changes. The most frequent changes are small, and large changes from which new forms emerge are far less frequent. The form of a hill, sand dune or accumulated snow, for example, is a balance between the resistance of the grains and the force of gravity; that balance between forces produces the angle of the slope. If more mass accumulates, for example, by 'saltating' sand grains, or by fresh snowfall higher up the slope, the slope will be in a state of 'self organised criticality',¹² poised close to the threshold of change. The addition of more grain may be enough to cause the form to change. The change may be small, just a few grains slipping down the slope, or more rarely it may be large, a catastrophic collapse over a large area. There may be more than one 'variable' that can induce change, for example, when water infiltrates the slope or it absorbs the energy of vibrations from earthquakes. Many other landforms develop to this condition, poised close to instability.¹³ Critical thresholds are those where the effect produced by a small change in one or more variables, such as the addition of a few grains of material, is disproportionately large or 'non-linear', producing a rapid and substantial collapse and a subsequent cascade of other changes, or a reorganisation from which new forms emerge. Unlike singular and small changes, the large and multiple changes produced by transition through critical threshold are irreversible. There are many continental scale ecological and climatic systems that are known to be sensitive to temperature changes and are

either close to their critical threshold and change now, or very soon will be as the world continues to warm.¹⁴ The transition of one or more of them through their critical threshold will provide positive feedback that has the potential to initiate a cascade of changes in others.

The Arctic summer sea ice, diminished by warmer air close to the surface, has a temperature threshold between 0.5° and 2° Celsius of global warming, and the timescale of complete transition through the threshold to an ice-free Arctic sea is thought to be very rapid, perhaps less than 10 years. There are many measured observations that suggest this threshold is very close. Transition will amplify warming and accelerate changes in the ecological systems of the high latitudes. The great ice sheets have such enormous volumes of ice that their complete transitions are widely thought to be slow, although there are many observations that strongly suggest that these transitions have begun, and are proceeding more rapidly than had been anticipated. The Greenland ice sheet has a temperature threshold between 1° and 2° Celsius of global warming, with a much longer timescale of complete transition of more than three centuries. When the transition is completed sea level will be 7 metres higher than it is today. The West Antarctic ice sheet has a temperature threshold of 3° Celsius or more of global warming, with a similar timescale of complete transition of more than three centuries. Complete transition will add a further sea level rise of 5 metres.

The Atlantic thermohaline overturning circulation is sensitive to increased freshwater entering the North Atlantic from the melting ice, with a temperature threshold of 3° Celsius or more of global warming. The reduction and eventual failure of the circulation will significantly shift the converge zones of the atmospheric circulation, inducing larger shifts to the existing climatic regimes across the Northern Hemisphere. The timescale of complete transition is thought to be about 100 years. The El Niño–Southern Oscillation is also similarly sensitive to temperatures of 3° Celsius or more of global warming, and will be greatly amplified after complete transition of about 100 years. The effects will be prolonged and see intense droughts in South-East Asia and other regions in similar latitudes. The monsoon belt has always been prone to abrupt changes, typically taking only one or two years for transition through its critical threshold, and the effects are prolonged droughts across much of Asia. Changes in the albedo of the surface of the Indian subcontinent are thought to be a trigger, but the history of rapid changes suggests that there are other variables that can drive this particular system through its critical threshold.

The forests of the world are conventionally thought to be sensitive to a temperature threshold of 3° Celsius or more of global warming, with a timescale for complete transition of less than 50 years. The shrinkage of spatial extent of the rainforests, including the Amazon, will diminish the transpiration of water and so decrease rainfall, change the albedo of the surface, induce the loss of other living species, and modify the exchange of oxygen and carbon dioxide between the atmosphere and the surface. The boreal forests are also thought to have a similar temperature sensitivity and a 50 year timescale for complete transition to a new form of ecological system such as grassland and shrubs. However, there are multiple variables that can increase that sensitivity and accelerate the transition to complete change of form, including drought, insect and viral infestations, and human predation. Other systems, particularly the tundra and permafrost, are observed and measured to be already in transition to new forms. It is not known how long the complete transitions will take, but the conventional assumption of 100 years now appears to be optimistic.

The increase in temperature observed in recent decades over Alaska provides a canonical example of the transition of a whole ecological system through a cascade of critical thresholds. The warming of the atmosphere has caused the snow to melt earlier in the spring, reduced the extent of sea ice, and increased the duration and depth of the summer melting of permafrost. Each effect provides a positive feedback that adds more warming to the atmosphere and amplifies the other effects. Substantial changes are now observed in the expansion of wetlands, changes to the flora and fauna of the tundra, in the general advance to the north of the tree line of forests, and the significant increases in the frequency and spatial extent of insect infestations over large areas of southern and central Alaska. One beetle, the spruce bark beetle, completed its life cycle in one year rather than in the two years that it took to do so in colder conditions. Spruce forests were already weakened by nine successive years of drought, and so highly vulnerable to the massive increase in insect infestations. Over large areas of south and central Alaska the spruce forests have died off, and the dead trees provide excellent fuel for wild fires, now increased in frequency and spatial extent.¹⁵ As the forests die back in the south and central areas of Alaska, shrubs and grasses replace them, and that in turn produces changes in the species of birds and animals that can live within the ecological system, and the density of their populations.

CLIMATE CHANGE

There have been many changes to the climate system in the past, at a variety of spatial and temporal scales. The climate is changing now and will continue to do so, although there are many uncertainties about the pace and the amplitude of the changes to come. Evidence from the past, and computational models and simulations that are based on that historical data, indicate that there are a large number of subsidiary systems that are close to or approaching their critical threshold. However, it is clear from current observations and measurements that the pace and amplitude of changes has been far more rapid than predicted. It is widely thought that there will continue to be an increase in the emission of greenhouse gases from the burning of fossil fuels in the coming decades, although it is not clear how big that increase will be. Even if it were possible to immediately eliminate any further increases, and to begin to make steady reductions in the quantity of gases emitted, the atmosphere will still retain the gases that have already been emitted, and so the world will continue to warm for many centuries.¹⁶ Evidence from the past suggests that the sensitivity of the climate to greenhouse gases may be variable rather than static, so that changes amplify exponentially as the climate warms. The metasystem is delicately poised, with many different critical thresholds across multiple scales of distance and time, so that a small stimulus at one scale may drive a great and rapid change at another.

In the last decade the differing projections of the amplitude of future surface warming above the current average world temperatures by the end of this century were synthesised in the report by the Intergovernmental Panel on Climate Change (IPPC), with a range of estimates of an increase from below 2° Celsius to above 6° Celsius.¹⁷ The climate simulations were run for different emission scenarios, either assuming that human activities would continue unchanged (the 'business-as-usual' scenario), or aimed at the stabilisation of emissions at a particular level, but with no probabilities attached to any of the scenarios. The higher projections were based on the assumption that the current failure to make any reductions at all in the emission of greenhouse gases will continue, and that emissions will continue to increase as the human population increases and fast feedbacks trigger a cascade of changes. Low estimates were based on the assumption that emissions will continue to rise only for the next decade or less, and then they will be reduced each year by 3 per cent. The middle of this range is based on the premise that emissions will continue to rise for one generation or 25 years before 'stabilising', and that steady reductions of 2 to 3 per cent per year will be achieved thereafter. There is an inherent optimism even in this projection,

as it is based on the assumptions that there will be widespread coordination of effective actions across the world, that future technical advances will make the 'stabilisation' and subsequent reductions in emissions possible in only one generation. However, despite the many initiatives and commitments to reducing emissions, there continues to be a steady increase in emissions of between 2 and 3 per cent each year and a consequent rise in the quantity of heat-retaining greenhouse gases in the atmosphere. On this evidence it is reasonable to assume that a large swing from a steady 3 per cent increase to a steady 3 per cent decrease in emissions in such a short time is extremely unlikely.

The projections from the 'MIT Integrated Global System Model' forecast that the median surface warming in 2091 to 2100 will be between 4.1° Celsius and 5.1° Celsius.¹⁸ The projections are 'probabilistic' rather than the scenario base adopted by the IPPC (a more formal mathematical approach to uncertainties), and the model has a higher resolution, and a more detailed representation of human activities across the world than any previous models. Furthermore, the model includes known fast and slow 'feedbacks' that inhibit or accelerate the processes of the climate metasystem, and the cascading effects of successive transitions through different critical thresholds. Significantly, the projections specifically exclude assumptions of future reductions in emissions by international policy changes, but do include mathematical models of irregular regional expansions and contractions of human activities abstracted from historical evidence.

As the percentage of carbon dioxide, nitrous oxide, methane and black carbon in the atmosphere continues to increase, so more heat energy can be absorbed and retained by the atmosphere, and the whole hydrological cycle will be accelerated. Evaporation will increase, and the patterns of atmospheric circulation, winds and precipitation will shift. More heat in the atmosphere will warm the surface below – a positive feedback that will continually increase the temperature of the atmosphere. The mid range projection of 4.6° Celsius global temperature increase will not be the end point of the warming, and it is far from certain that the convention of a century-long timescale is accurate. The evidence from the past shows that large climate changes have frequently occurred far more rapidly. It has long been thought that the thermal inertia of the oceans and the great ice sheets of Greenland and Antarctica would delay the onset of the effects of the current warming by centuries. However, the accumulation of evidence from ice cores, tree rings, corals and sedimentary deposits in lakes and oceans, indicates many episodes of abrupt climate changes in the past, including

some with a rapid onset of between three years to a decade.¹⁹ Positive feedbacks from the dynamical processes of the land, oceans and ice sheets, and from the effects of the extended metabolism of civilisation, interact with and accelerate each other. Positive feedbacks accelerate warming²⁰ and can induce very large and rapid fluctuations in the climate metasystem.²¹

The increase of surface air temperatures is conventionally expressed as an average across the world, but the pattern is differentiated, with smaller temperature rises near to the equator, much larger rises nearer the poles, and more rapid increases over the interior of continents than over the oceans. Temperature increases are also much greater, and more rapid, at the high altitudes in the mountain ranges of all latitudes. The observed shrinking of the permafrost region in recent years is well documented, and extensive areas of Canada, Alaska and Siberia will have increasingly unstable soils as the melt accelerates. The release of carbon dioxide and methane from the newly warmed ground will add yet more greenhouse gases to the atmosphere, providing another positive feedback. The increased air temperature in the Arctic region in recent years²² has progressively reduced the geographical area and the thickness of sea ice, with increasingly large areas of open water in recent summers. Winter, spring and autumn temperatures in recent years have also been much warmer than the general pattern of the 20th century, with earlier melting in the spring and later formation of sea ice in the autumn and winter.²³ Right across the Northern Hemisphere snowfall will continue to decrease in the coming decades, and the flow of freshwater into the Arctic Ocean from the melting glaciers will accelerate further.

The great ice sheets are now melting at a faster rate than was ever anticipated, and the subsequent increase in albedo change generates a strong positive feedback that will further accelerate the ice melt until the physical stability of the ice sheets is undermined. Once this critical threshold is reached, rapid disintegration and break up will follow. It has long been recognised that a global temperature increase of just 3° Celsius will eventually lead to the elimination of the Greenland ice sheet, and up to 7 metres of sea level rise,²⁴ but the pace of recent changes in the ice sheets and in glaciers suggests that the predicted rise in sea level during this century has been substantially underestimated. It is now thought that if all the positive feedbacks continue to accelerate, sea level rise of up to 2 metres could occur by the end of the century, although a 1 to 1.5 metre rise is currently considered to be more likely.²⁵

The very large river delta regions of Asia, including the Huang He, Changjiang (Yangtze), Pearl, Red, Mekong, Chao Phraya, Irrawaddy, Ganges-Brahmaputra, and Indus rivers have a total area of almost one million square kilometres, all of which is likely to be completely inundated within this century. Across the world almost all of the largest cities are located on rivers and coastal plains, with extensive land areas and infrastructure at or below 2 metres elevation, including London, Venice, New York, Boston, Miami, New Orleans, Los Angeles, Mumbai, Shanghai, Tokyo, Lagos and Capetown. All are highly vulnerable to sea level rise, and to the increased intensity and frequency of large storm surges in a warming world. Rising sea levels will disrupt the flow of freshwater in river deltas and the salt water will contaminate estuaries and groundwater. Increased salinisation will reduce the freshwater available for regional ecological systems, for agriculture and for cities.

As the glaciers continue to shrink, and snowfalls lessen, the quantity of freshwater they supply to river systems is reduced. The glaciers and snow packs of the mid latitude mountain ranges along the Pacific coast of North and South America, of the Alps, Pyrenees and Himalayas, and in New Zealand have all been in retreat for several decades, and the observed changes in ice melting and water run-off will accelerate further during the coming century. The major river systems that they supply will have increasingly early peak flows in winter and spring and greatly reduced summer flows when the demand for agricultural irrigation and drinking water is at its greatest.²⁶ The inevitable outcome is that these rivers will have little or no flow in the summer months, and by mid century it is anticipated that in many locations there will be complete failure of all river systems and lakes that are fed by mountain glaciers and snow packs.

As the world continues to warm in the coming decades, it is anticipated that the additional heat energy retained in the atmosphere will induce further and more pronounced changes in the mid latitudes 'inter-tropical convergence' zones. This is where warm, moist air rises up into the atmospheric convection cells, and they will shift substantially towards the poles in both hemispheres. Changes in the geographical distribution, variation and intensity of rainfall and monsoons around the world will follow, with complex regional and local outcomes. The intensity and frequency of extreme rainfall and storm events will further increase in regions that are already experiencing extreme rainfall events, including the west coast of India, north-east India, the Bay of Bengal and Bangladesh. There will also be an intensification of drought in the latitudinal band of deserts, which

will expand their geographical area substantially, accelerating the observed drying out and desertification in these latitudes.²⁷ The most severe and prolonged droughts will occur in the south-western region of North America,²⁸ in the central region of South America, in South Africa and southern Australia, in the northern and western Mediterranean region and in the Sahara. In Europe, the reduction of snowfall in the Alps and the earlier melting of the snow will initially increase the flow of the Rhine and other rivers dramatically in the spring, with risks of widespread flooding across Germany and Holland, but it will also substantially reduce water flow by more than half in the summer months. On the Mediterranean side of the Alps, the land will become increasingly arid from the lack of water and very high temperatures, and desertification will accelerate. Forests will die back in Europe and Asia, and the northern tree line will shift much further into the high latitudes that are currently semiarid savannahs or tundra. In North and South America there will be extensive loss of forest areas as temperatures rise and freshwater diminishes.²⁹ Agricultural cultivation, particularly wheat, rice and maize, will become increasingly difficult and likely to fail in all tropical regions, where crops are already close to their thermal threshold. As temperatures continue to rise, lack of water will also affect cultivation in the more temperate mid-latitudes, where it is anticipated that crop yields will eventually fall after an initial increase. Forests will shrink back and become increasingly vulnerable to fire and pests as droughts become more intense and frequent in subtropical and southern temperate latitudes, especially in the western United States, northern China, southern Europe, the Mediterranean and Australia. Fuel, charcoal and other forest products that are essential for a significant fraction of the world's rural populations will be rapidly diminished. Many forest ecological systems have been affected by climate changes in recent decades, and those effects will be accelerated as the climate continues to warm.

COLLAPSE AND REORGANISATION

Despite their many variations, there are some common characteristics that all the systems of civilisation have exhibited when close to their particular critical threshold, and so vulnerable to change. These include accelerating informational complexity, rising energetic costs of the information and processes that collect and regulate the flow of energy and material over extended distances, small but increasingly frequent oscillations between expansion and contraction, diminishing returns and rising costs, expanding populations and disproportionate and inequitable distribution of consumption. Climatic and ecological changes have often initiated decline, and the responses to diminishing

returns and decline have included warfare to capture or defend resources, inappropriate intensification of agricultural and energetic resources already at maximum capacity, planned simplification, and abandonment and mass migrations. These processes have usually occurred over the course of a single human generation, but some persisted over several generations.

It is clear that the current metabolic system of the world, with its accelerating informational complexity, extreme velocity and volumes of fuel and food energy flowing across continents and oceans, and high but inequitable energy and material consumption, exhibits similar characteristics. Despite the high density flow of information and the high proportion of the population engaged in regulating and administering the energy and material flows, the global system is now highly vulnerable to disruption and perturbation. Further intensification of flows through the extended metabolism, sometimes referred to as 'continuing economic growth', is incompatible with the significant reductions in agricultural returns of food and fuel energy that are anticipated as the climate changes over the next decades. It follows that if the predicted increase in the population of the world, from significantly more than 6 billion today to over 9 billion in the next three decades, were to come about it would precipitate the catastrophic collapse of the system. A decade ago there were 450 cities in the world with a population of 1 million or more. An additional 3 billion people will require the equivalent of 3,000 new cities of that size to be built within one generation. The quantities of materials required, and the energy required to transform and manipulate those materials and construct the cities are immense. The supply of food and fuel energy to support an additional population half the size again of the world's existing population cannot be met by the current metabolic system that is already operating close to its maximum capacity.

The outcomes of the collapse of cultural systems in the past have been strikingly similar to the collapse and reorganisation of natural complex systems; the complete loss of social order and dispersal of the people by famine, disease, abandonment of cities and mass migration; a regrouping of the cities and settlements within the extended systems into smaller and simpler assemblies with reduced connections to other local and regional systems or a reorganisation to a lower level of complexity; and the reordering of the cities, people and subsidiary systems into a more integrated assembly or more complex reorganisation with a higher flow of energy and information. If it is so that the world system is now poised close to its critical threshold of stability, and if it is

so that the climate changes currently under way cannot be inhibited by significant reduction in emissions within the next decade, then it follows that changes will cascade through all the systems of nature and civilisation. Although the scale of the world system is currently at least one order of magnitude greater than the largest imperial system of the past,³⁰ there are sufficient similarities to suggest the delineation of three principal modes of transition through the critical thresholds of nature and civilisation from which new forms will emerge.

DISPERSAL AND ABANDONMENT

If the present course is continued, the quantities of resources invested in the intensification of flows of energy, information and materials through the system will necessarily increase as the population rises. Emissions will be accelerated by the intensification of the system and resources will be further depleted. The inequitable differentiation of resources and consumption within nations and between them will continue, the existing social and commercial institutions will be maintained and populations will continue to expand exponentially. Thousands of new cities need be built, but are likely to be significantly underfunded, and in consequence poorly constructed and inadequately serviced as resources become increasingly scarce. The widespread failure to recognise the necessity for a timescale of two generations and massive resource commitment for continental scale deployment and activation of new energy systems will continue to inhibit the funding of research and development. Similarly, there is a widespread failure to recognise that the first regional scale transitions to new ecological systems have already commenced and are likely to be completed within two generations. Significant reductions in agricultural returns and freshwater will accelerate rapidly over the next generation, and it is very likely that the differentiated and inequitable distribution will lead to widespread social unrest and famine in the most ecologically stressed regions. In these regions construction of new buildings and cities will cease altogether, and it is likely that within one generation large territories will be abandoned. Mass migration into territories that are less stressed is likely to be resisted by closed borders and by arms. In the less stressed regions, despite rapidly diminishing ecological resources, intensification will have to be continued. This will provide positive feedback that will amplify and accelerate climate and ecological changes.

The transoceanic and intercontinental flow of energy, materials and people is likely to be significantly diminished, and countries and regions that import a high percentage of their food and fuel energy from outside their borders will be

greatly stressed. As the climatic and ecological transitions accelerate, widespread social unrest, famine, and sporadic but massive incursions will lead continental scale systems with substantial agricultural and energy resources within their borders to seal their boundaries entirely. It is likely that their own diminished resources will make it increasingly difficult to feed and fuel their own system, and they are likely to cease transoceanic and intercontinental flows altogether. Sealed within their own borders they may well allow the depredations of famine and disease to reduce their populations in order to husband their resources. Rising sea level, famine and disease will eventually lead to the widespread abandonment of delta cities, and the mass dispersals of people will further stress immediately adjacent territories. The smaller systems and aggregations of small systems will be weakened by mass famine and increasingly limited resources and will then rapidly decline and effectively cease to exist as distinct entities.

Widespread abandonment of cities in significantly stressed territories will follow, but the lack of energy resources, insufficient land for subsistence agriculture and the sealed borders of larger systems will rapidly reduce the population in these territories. Isolated 'refugia' will certainly continue at a variety of scales and in locations where the climatic and ecological changes are broadly favourable or at least endurable, but the world population will inevitably be significantly reduced. Within four generations the extended metabolic system of civilisation that now extends across the world will have shrunk back to a few strictly contained metabolic systems with defined boundaries and restricted connections to other systems. The first climatic and ecological transitions will then be completed and changes will have begun to cascade rapidly through both natural and cultural systems. New forms of ecological system will have emerged in the high and mid latitudes, the deserts will be expanding and sea level will have risen. Some 'refugia' may gradually decay and decline as their existing energy, agricultural and information systems fail over time. It is likely, however, that the majority of 'refugia' cities and their localised metabolic systems will have already begun to change, rebuilding themselves slowly cell by cell as resources allow, and new forms of metabolic system based on local resources will emerge. New building and ultimately new city forms will emerge over multiple generations. It is likely that the warming of the climate will continue over the next millennia, and the evolution of the forms and metabolic systems of settlements, cities and systems of cities that are viable in those new conditions from the small base of isolated 'refugia' into an integrated new world system is likely to occur at best very slowly, if at all.

SIMPLIFICATION

If it is so that a worldwide consensus does eventually emerge that the present course cannot be continued and that consumption of all kinds should be reduced, it is likely to take a whole human generation to do so. Similarly, a widespread agreement to limit the expansion of the population is also likely to take a generation to become effective, and in the meantime population growth will continue but at a slower rate than is currently predicted. The quantities of resources invested in the intensification of flows of energy, information and materials through the system will be slowly reduced, although the increase in population will inhibit the reduction of flows for at least one generation. Some elements within all societies may never abandon their commitment to high energy systems, expansion and increasing and inequitable consumption. Those elements will eventually be coerced, or overtaken by events, but their resistance to change will be effective for at least one generation. Emissions will continue to increase throughout the next generation but at a slower rate, and it is likely that emissions will only begin to significantly diminish after two generations have passed. The inequitable differentiation of resources and consumption within nations and between them will be minimally moderated by the gradual reduction in consumption, and existing social and commercial institutions will slowly be reduced and transformed to align with the new consensus. Research and development of new sources of energy will continue to be minimally funded, but will slowly increase as recognition of the two generation-timescale and resource commitment necessary for continental scale deployment and activation of new energy systems becomes widespread. Deployment of solar energy systems and other energy sources that do not emit greenhouse gases and black carbon particles may provide one third of all fuel energy within two generations, and two thirds within three generations.

Significant reductions in agricultural returns and freshwater will accelerate rapidly over the next two generations, and the expanded populations will reduce surplus capacity everywhere. Despite willingness to assist others, it is likely that there will be insufficient surplus to ameliorate the famine and widespread social distress in the most ecologically stressed regions. In these regions construction of new buildings and cities will cease altogether, and city after city will be abandoned. Mass migration into territories that are less ecologically stressed will be restricted, but not completely resisted until the numbers threaten to overwhelm their own capacity. Ultimately there will be too many migrants and borders are likely to be closed. Reduced consumption everywhere will begin to

reduce transoceanic and intercontinental flows within two generations, but it is the shortage of fuel and food supplies for the expanded population that is most likely to significantly reduce them. As the climatic and ecological transitions accelerate, widespread social unrest, famine, and sporadic but massive incursions will increase the necessity for smaller systems and assemblies of smaller systems to close their borders and defend what resources they have to sustain their populations at a low level of consumption. The majority of small systems are currently located within regions that have a high dependency on fuel and food imports from across the world, and widespread abandonment of their cities will follow the significantly diminished flow within two generations. If transoceanic and intercontinental flows do eventually cease altogether, the majority of the population in ecologically distressed territories will attempt to migrate or will succumb to famine.

Rising sea level, famine and disease will lead to the widespread abandonment of delta cities, and the mass dispersals of peoples will further stress immediately adjacent territories. Continental scale systems with expanded populations and diminished resources will only be able to accept large numbers of migrants by further reducing consumption. Within their own territories the migration from cities to smaller more rural cities and settlements will accelerate over two generations, and within three generations even the largest cities are very likely to have significantly reduced populations. Societal commitment to reducing the population will inevitably be reinforced by coercion and closed borders, and will only begin to be significantly effective within three generations. The new forms of ecological system in the high and mid latitudes, and the expanding deserts and rising sea level will produce significant stress on the emerging reconfiguration of the larger systems. Local and regional scale patches will decline and fail as their local ecological systems change and further reduce agricultural returns. Famine will further reduce the population. Within four generations the reduced population will be widely dispersed in localised systems of small cities and settlements, with the major fraction of their food and fuel energy extracted from their local environment. Connections to other systems will be maintained at a minimal level. Over multiple generations the reconfiguration of agricultural and low energy systems will be refined and fully deployed, regional connections will slowly intensify and a low energy and low density agrarian system with new forms of buildings and cities and a network topology adapted to the new forms of ecological system and climatic regime will emerge.

HIGHER COMPLEXITY

If it is so that civilisation will transit through multiple critical thresholds to higher complexity, those elements within all societies that are unwilling or unable to abandon their commitment to inequitable consumption will have to be bypassed within the next generation. The flows of energy, information and materials through the system will have to be accelerated. Despite the widespread consensus that there is insufficient capacity in the world metabolic system to successfully transit through the multiple critical thresholds, population growth is likely to continue unabated, necessitating extensive building of new cities and increased density in some existing cities. The increase of emissions caused by the expansion of populations, the mass construction of new cities and the intensification of energy, information and material flows will be amplified by the ecological transitions induced by climate change, and initiate the cascade of multiple transitions in all ecological systems. Thereafter, widespread observation that the impending cascade of transitions through all the systems of nature and civilisation is under way and accelerating will lead to societal commitment to the redesign and reconfiguration of the world's metabolic system.

The development of new sources of energy that do not emit greenhouse gases and black carbon particles will require the investment of a significant fraction of the global economy, as will the accelerated flow of information. The widespread deployment of new energy systems in suitable areas to supply half of all the fuel energy of the world is likely to take two complete generations, and full deployment in those territories is unlikely to be achieved before three generations have passed. Voluntary commitment to limiting the expansion of the population may begin to slow the rate of expansion within one generation, but will have to be reinforced by strong societal commitment with some coercion if the world population is to be stabilised within two generations. Emissions will thus continue to increase, and are unlikely to begin to diminish until the deployment of new energy systems has reached half of all energy generation. Investment in information, both research and distribution systems, is likely to continue at current levels and is only likely to increase when the first regional scale ecological transitions begin to significantly diminish agricultural returns and reduce freshwater supplies, and all resources become scarce. The wide dissemination of free information to all parts of the world will then begin to have a significant impact on all energy and material transformations, and the transition to a truly 'distributed intelligence' world system will be accelerated. The first regional scale transitions to new ecological systems are likely to be

completed within the next two generations, and severe famines in ecologically stressed regions will lead to widespread social unrest in those territories. In those circumstances the construction of new buildings and cities will rapidly diminish and ultimately cease altogether. In city after city the generation-long process of abandonment and mass migrations will commence.

Completed ecological transitions of the high and mid latitudes, expanded deserts and rising sea level will produce significant stress on all systems, and there will be multiple local and some regional failures everywhere. However, high density energy and information systems will help to ameliorate a significant fraction of the difficulties in providing food and fuel that all systems will experience. Systemic agricultural failure and the depredations of famine and disease in regions that have not adapted their systems will induce widespread abandonment of their cities in favour of small subsistence level rural systems where there is available territory, and mass migrations will increase the stresses elsewhere. There is very likely to be a significant reduction in the world population within four generations.

Higher complexity in all systems is characterised by increased differentiation and increased connections, in conjunction with an accelerated flow of energy. In the systems of civilisation the accelerated flow of energy has always been coupled to increased quantities and kinds of information and to greater transformation and flow of materials. In each of the newly emerged and reconfigured ecological systems, new forms of cities will begin to emerge within two generations, and be widely proliferated within four generations. Continental flows of energy and material are likely to be intensified as intelligent inhabited infrastructures that unite and service ecological systems rather than divide them come on-line. Transoceanic flows are only likely to be significantly reduced within three generations if the flow of information accelerates and begins to reduce the necessity to move transformed materials over great distances.

In the high latitudes where the tundra and permafrost will have completed transition to boreal forests and grassland, new large but low density cities will emerge that are organised as intelligent ecological infrastructures that capture methane and sequester carbon, fix nitrogen, collect and purify water, and produce agriculture materials that are symbiotically integrated with the regional ecological system. Surface farming of methane released by the melting permafrost will provide a substantial high-density energy source for

systems of cities above the soft terrain right across the high latitudes, incorporating large-scale intensive agricultural systems. These city systems will be morphologically and topologically very different from cities today, most likely configured with multiple small nuclei within networks of inhabited energy and agricultural infrastructures.

In the mid latitudes new agricultural infrastructures will be inhabited more densely and large urban densely inhabited systems will be gradually transformed as their populations decline, but these transformations will be very slow over multiple generations. Cities and metabolic systems everywhere will be more differentiated in scale, intensively but selectively connected in new network-topologies that are symbiotically configured to the flows in their local and regional ecological systems. Within three generations rising sea level will have a profound effect on river delta cities and densely inhabited low-lying coastal territories. Some delta cities will gradually be transformed to highly engineered and intelligent polders, substantially reconfigured above and below sea level as marine ecological machines that capture energy and have extensive breeding and husbandry of marine life. However, where such transformations fail from lack of investment and political will, rising sea levels, famine and disease will induce widespread abandonment of most if not all delta cities and many coastal territories, and the mass dispersals of peoples will further stress immediately adjacent inland territories.

Desert cities will be gradually transformed into inhabited energy infrastructures, with vast arrays of solar energy systems generating constant supplies of high-density energy that can be transmitted across continental scale distances through intelligent electrical grids.

It is clear that the world is within the horizon of a systemic change, and that transitions through multiple critical thresholds will cascade through all the systems of nature and civilisation. New forms will emerge down through all the generations to come, and they will develop with new connections between them as they proliferate across the surface of the earth.

1 One human generation = 25 years.

2 Moses, ME, S Forrest, AL Davis, MA Lodder and JH Brown, 'Scaling Theory for Information Networks', *Journal of The Royal Society Interface*, vol 5, no 29, 2008, pp 1469–80.

3 Barabási, AL and R Albert, 'Emergence of Scaling in Random Networks', *Science*, vol 286, 1999, pp 509–12; and Jeong, H, B Tombor, R Albert, ZN Oltvai and AL Barabási, 'The Large-Scale Organization of Metabolic Networks', *Nature*, vol 407, 2000, pp 651–4.

- 4 Data from *The World at Six Billion*, Population Division, Department of Economic and Social Affairs, United Nations Secretariat, 1999.
- 5 The larger the number gets, the faster it grows.
- 6 By 2050. The US Census Bureau International Data Base, April 2009.
- 7 Animals weighing more than 44 kilograms are generally referred to as 'megafauna' – but the mammoths and mastodons weighed over 1,000 kilograms.
- 8 Hope, G, 'Early Fire and Forest Change in the Baliem Valley, Irian Jaya, Indonesia', *Journal of Biogeography*, vol 25, 1998, pp 453–61.
- 9 Ledig, FT, 'Human Impacts on Genetic Diversity in Forest Ecosystems', *Oikos*, vol 63, 1992, pp 87–108.
- 10 Foley, JA, C Monfreda, N Ramankutty and D Zaks, 'Our Share of the Planetary Pie', *Proceedings of the National Academy of Sciences*, vol 104, 2007, pp 12585–6.
- 11 Robock, A and HF Graf, 'Effects of Pre-Industrial Human Activities on Climate', *Chemosphere*, vol 5, 1994, pp 1087–97.
- 12 Bak, P, C Tang and K Wiesenfeld, 'Self-Organized Criticality: an explanation of the 1/f noise', *Physical Review Letters*, vol 59, 1987, pp 381–4. Presents the first argument that unconstrained dynamical systems naturally evolve into a self-organised critical state.
- 13 Schumm, SA, 'Geomorphic Thresholds: the concept and its applications', *Transactions of the Institute of British Geographers*, New Series vol 4, 1979, pp 485–515.
- 14 Lenton, TM, H Held, E Kriegler, JW Hall, W Lucht, S Rahmstorf and HJ Schellnhuber, 'Tipping Elements in the Earth's Climate System', *Proceedings of the National Academy of Sciences*, vol 105, no 6, 2008, pp 1786–93.
- 15 'Thresholds of Climate Change in Ecosystems', US Climate Change Science Program and the Subcommittee on Global Change Research, 2009, pp 4, 32–6.
- 16 'There is high agreement and much evidence that with current climate change mitigation policies and related sustainable development practices, global GHG emissions will continue to grow over the next few decades'; and 'Both past and future anthropogenic CO₂ emissions will continue to contribute to warming and sea level rise for more than a thousand years'. *Climate Change 2007 Synthesis Report*, Intergovernmental Panel on Climate Change, 2008.
- 17 '... the best estimate for the low scenario is 1.8°C (likely range is 1.1°C to 2.9°C), and the best estimate for the high scenario is 4.0°C (likely range is 2.4°C to 6.4°C)'. *Climate Change 2007 Synthesis Report*, Intergovernmental Panel on Climate Change, 2008.
- 18 Sokolov, AP, PH Stone, CE Forest, R Prinn, MC Sarofim, M Webster, S Paltsev, CA Schlosser, D Kicklighter, S Dutkiewicz, J Reilly, C Wang, B Felzer and HD Jacoby, 'Probabilistic Forecast for 21st Century Climate Based on Uncertainties in Emissions (without Policy) and Climate Parameters', *MIT Joint Program on the Science and Policy of Global Change*, 2009.
- 19 Jones, PD, KR Briffa, TP Barnett and SFB Tett, 'High-Resolution Palaeoclimatic Records for the Last Millennium: interpretation, integration and comparison with General Circulation Model control-run temperatures', *The Holocene*, vol 8, 1998, pp 455–71; and Mann, ME, RS Bradley and MK Hughes, 'Global-Scale Temperature Patterns and Climate Forcing over the Past Six Centuries', *Nature*, vol 392, 1998, pp 779–87; and Mann, ME, RS Bradley and MK Hughes, 'Northern Hemisphere Temperatures during the Past Millennium: inferences, uncertainties and limitations', *Geophysical Research Letters*, vol 26, 1999, pp 759–62.
- 20 Cox, PM, RA Betts, CD Jones, SA Spall and J Totterdell, 'Acceleration of Global Warming due to Carbon-Cycle Feedbacks in a Coupled Climate Model', *Nature*, vol 408, 2000, pp 184–7.
- 21 Hansen, James, Makiko Sato, Pushker Kharecha, Gary Russell, David W Lea and Mark Siddall, 'Climate Change and Trace Gases', *Philosophical Transactions of the Royal Society of London A*, vol 365, 2007, pp 1925–54.
- 22 'Arctic Report 2008 – Atmosphere', *US National Oceanic and Atmospheric Administration*, <http://www.arctic.noaa.gov/reportcard/atmosphere.html>.
- 23 Overland, JE, M Wang, and S Salo, 'The Recent Arctic Warm Period', *Tellus*, vol 60A, 2008, pp 589–97.

- 24 *Climate Change 2007 Synthesis Report*, Intergovernmental Panel on Climate Change, 2008.
- 25 Pfeffer, WT, JT Harper and S O'Neel, 'Kinematic Constraints on Glacier Contributions to 21st-Century Sea-Level Rise', *Science*, vol 321, 2008, pp 1340–3.
- 26 Barnett, TP, JC Adam and DP Lettenmaier, 'Potential Impacts of a Warming Climate on Water Availability in Snow-Dominated Regions', *Nature*, vol 438, 2005, pp 303–9.
- 27 'Globally, the area of land classified as very dry has more than doubled since 1970', *Climate Change 2007 Synthesis Report*, Intergovernmental Panel on Climate Change, 2008.
- 28 Cook, ER, R Seager, MA Cane and DW Stahle, 'North American Drought: reconstructions, causes, and consequences', *Earth Science Reviews*, vol 81, 2007, pp 93–134.
- 29 Scholze, M, W Knorr, NW Arnell and CI Prentice, 'A Climate-Change Risk Analysis for World Ecosystems', *Proceedings of the National Academy of Sciences*, vol 103, 2006, pp 13116–20.
- 30 In 1938 the British Empire is thought to have had a population of over 500 million or half a billion. In 2009 the world population is thought to be rapidly approaching 7 billion.